

Appearance-Independent Tactile Design Using Different Embroidery Stitches

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Abstract

We present a novel method to enhance the tactile properties of fabric interfaces through embroidery, enabling appearance-independent tactile design. Unlike existing approaches that alter geometric properties, such as shape or contours, our method uses commercially available embroidery techniques to achieve tactile variations while preserving the visual consistency of the fabric. By adjusting parameters like stitch length, density, and deviation, we establish a design space for reproducible and tactilely diverse embroidery patterns. Our user study, involving 26 embroidered samples, demonstrates how different stitching techniques influence tactile perception, including qualities like smoothness, roughness, and directional guidance. Attendees will experience several use cases at the demonstration, such as eyes-free interfaces for sliders and directional keypads, tactile learning tools for children, and assistive applications for visually impaired individuals. This work highlights the potential of embroidery as a versatile and practical solution for designing tactilely engaging fabric interfaces.

CCS Concepts

• **Human-centered computing** → *Haptic devices*.

Keywords

tactile, fabric, embroidery, fabrication

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1 Introduction

The advancement of smart textile technologies has made it possible to integrate interactive functionalities into everyday fabric-based products. So far, research efforts have focused on technological aspects, such as sensing and fabrication methods. However, to create effective interactions based on the unique characteristics of fabrics, it is important to investigate not only the technical aspects, but also the material aspects of its *tactile* design.

Among these, the “tactile perception” of the fabric has gained considerable attention. For example, previous work proposed the design of eyes-free interfaces that would enable users to distinguish features solely by touch [3], while others explored the tactile perception of affordances that would guide specific movements [1]. However, these existing approaches require altering the geometric properties of the fabric interface, such as its shape [2–4] or contours [3, 4]. Although these methods effectively manipulate the tactile properties of the interface, they also alter its visual appearance. Methods that achieve desirable tactile properties without altering the appearance of fabric interfaces would thus contribute to maintaining the aesthetics of the interface.

To address this challenge, we investigate a novel approach to achieve tactile variations using different embroidery stitches. More specifically, we explore how the parameters available in commercial stitching software, such as stitch length and density (Figure 1), can lead to different tactile properties of embroidery without significantly altering its shape or size. This approach minimizes visual alteration, offering a widely applicable solution for fabric interfaces.

In this interactivity paper, we first describe the underlying concepts of embroidery design and how manipulating certain of its parameters can produce embroideries of different tactile textures, yet visually similar. In addition, we describe an evaluation of tactile perception by different stitching techniques. We then describe typical use cases that can leverage embroideries of different tactile design, which CHI attendees will be able to experience at the conference. Finally, we briefly reflect on the avenues for future research opened by this work before concluding.

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2 Embroidery Design

Given the extensive variety of embroidery stitching techniques, our work first aimed at narrowing down the options to identify a design space for stitching styles that can modify tactile properties without altering the appearance. In particular, general embroidery designers may find it challenging to use stitching techniques unavailable in common embroidery design software. Therefore, we decided to explore the potential set of stitching styles available in three major software platforms: Shishu Pro 11 (BROTHER INDUSTRIES, LTD.)¹, Wilcom EmbroideryStudio (Wilcom)², and DG16 (Tajima Software Solutions Inc.)³.

From the stitching styles commonly available in these software platforms, we excluded the following types, as they do not align with the focus of this study:

- Highly decorative motifs composed of collections of shapes such as hearts or leaves.
- Irregular stitching styles where parameters such as stitch length, angle, or density are inconsistent.
- Stitching styles with low stitch density, resulting in visible fabric beneath, such as cross-stitch or net fill.

We also conducted interviews with five experts in embroidery design, to gather opinions on which stitching styles to evaluate. Their expertise ranged from a few months to over 30 years of experience. We first explained the process used to create the potential set of stitching styles and the parameters included in the set. We then presented 56 different embroidery samples that the experts physically examined while responding to several questions. Subsequently, we asked them about the feasibility and reproducibility of the stitches included in the potential set.

Our interview participants identified two categories of stitches that would not necessarily be adapted: those that distorted the fabric (such as concentric or radial patterns), making it difficult to achieve consistent and aesthetically pleasing results, and those that would be financially too expensive to use, despite being feasible to execute. We therefore excluded these two categories of stitches.

We also inquired about ideas for tactilely impactful stitches that were not included in the provisional set. Among the suggestions from the experts, program-controlled fill stitches (which involve systematically varied pitch parameters) and 3D embroidery incorporating urethane stood out. These stitches were deemed feasible and highly reproducible using standard sewing machines, materials, and design software, and were therefore included in our approach. On the other hand, stitches requiring specialized machines or threads were excluded due to their limited applicability to generalizable knowledge.

In the end, we therefore test the following patterns (summarized Table 1 and whose properties are illustrated Figure 1). For a *Stitch Type* set to *Fill*, we test the 18 patterns resulting of the combinations of *LENGTH* (1, 4, or 8mm), *DENSITY* (2 or 6 threads/mm) and *DEVIATION* (0%, 25% or 50%). In addition, we test patterns using the *Radiation*, *Concentric Circular*, *Cross Stitch* and *3D fills* TYPES (see Table 1 for other parameters that were set based on pilot tests).

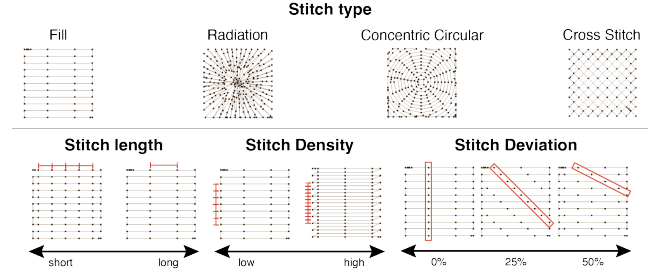


Figure 1: Stitch properties targeted in this paper. The black line is the thread and the black dot is the needle penetration point. Stitch length is the interval between the needle penetration points, stitch density is the spacing between the threads, and stitch deviation is the displacement of the needle penetration points.

Table 1: List of embroidery sample stitching properties(Figure 1).

ID	Stitch Type	Stitch Length	Stitch Density	Stitch Deviation
1	Fill	1mm	2 thread/mm	0%
2	Fill	1mm	2 thread/mm	25%
3	Fill	1mm	2 thread/mm	50%
4	Fill	1mm	6 thread/mm	0%
5	Fill	1mm	6 thread/mm	25%
6	Fill	1mm	6 thread/mm	50%
7	Fill	4mm	2 thread/mm	0%
8	Fill	4mm	2 thread/mm	25%
9	Fill	4mm	2 thread/mm	50%
10	Fill	4mm	6 thread/mm	0%
11	Fill	4mm	6 thread/mm	25%
12	Fill	4mm	6 thread/mm	50%
13	Fill	8mm	2 thread/mm	0%
14	Fill	8mm	2 thread/mm	25%
15	Fill	8mm	2 thread/mm	50%
16	Fill	8mm	6 thread/mm	0%
17	Fill	8mm	6 thread/mm	25%
18	Fill	8mm	6 thread/mm	50%
19	Radiation	4mm	2 thread/mm	-
20	Concentric Circular	4mm	2 thread/mm	-
21	Cross Stitch	6mm	1.5 thread/mm	-
22	3D Fill	4mm	6 thread/mm	0%
23	3D Fill	8mm	6 thread/mm	0%
24	Programed Fill (Higaki)	-	6 thread/mm	-
25	Programed Fill (Saya)	-	6 thread/mm	-
26	Programed Fill (Hishi)	-	6 thread/mm	-

3 Preliminary tactile perception experiment

3.1 Methodology

We recruited 12 participants (6 males, 6 females) aged 18 and above (10s: 2, 20s: 8, 30s: 1, and 40s: 1). The procedure consisted of the following steps:

- (1) Participants were asked to explore embroidered samples of the 26 selected patterns, presented in a randomized order to ensure exposure to a variety of textures. They were instructed to examine the embroidery surfaces with their fingers, employing different directions and speeds to fully experience the tactile variations.

¹<https://www.brother.co.jp/product/hsm/special/shishu-pro/index.aspx>

²<https://wilcom.com/embroiderystudio>

³<https://www.tajima.com/product/dg16/>

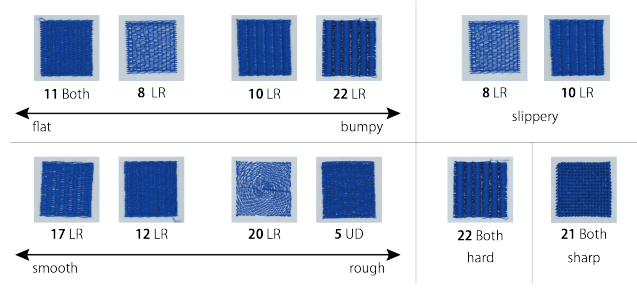


Figure 2: Embroideries with distinctive tactile perception based on participants responses. Numbers refer to the ID of the embroidery in 1; LR indicates the embroidery was perceived in the left/right direction, UD in the up/down direction, and Both in both directions.

- (2) In a subsequent task, participants touched the embroidery samples in a different randomized order and provided subjective ratings on the following tactile qualities: comfort, sharpness, continuity/discontinuity, flatness/unevenness, smoothness, warmth/coolness, hardness, and movement-guidance. The ratings were collected using 5-points Likert-type items. To investigate the anisotropy of tactile perception, most questions required responses for both up/down and right/left directions.

Embroidery samples were mounted in small wooden hoop, securely fixed within a frame. To prevent visual bias, participants' could not see the embroidery, ensuring they could only rely on tactile perception. Regardless of the embroidery style, all samples were standardized to a 2.5 cm square shape. To mitigate order effects, a balanced Latin square design was employed, ensuring that the sequence of presented samples varied across participants.

3.2 Result and Discussion

Figure 2 highlights embroidery patterns that produced distinctive results in the tactile perception questionnaire. For fill stitching (Patterns 1–18 (Table 1)), tactile properties such as the intensity of tactile stimulation, flatness/roughness, and smoothness/coarseness varied based on stitch length, stitch density, and deviation. In contrast, radial stitching (Pattern 19 (Table 1)) exhibited minimal distinctive features, while concentric stitching (Pattern 20 (Table 1)) was characterized as coarse, and cross-stitching (Pattern 21 (Table 1)) was perceived as sharp, coarse, and hard. 3D fill stitching (Patterns 22 and 23 (Table 1)), enhanced with polyurethane to emphasize surface irregularities, resulted in stronger perceptions of roughness. Programmed fill stitching (Patterns 24–26 (Table 1)) showed minimal differences in tactile perception, suggesting that intricate and complex patterns had little influence on tactile properties. Furthermore, differences in tactile perception between right/left and up/down directions were observed for many embroidery patterns, confirming the anisotropy of tactile perception.

Regarding finger movement guidance, most embroidery patterns, except for Patterns 19–21, exhibited directional guidance along the right/left direction. This is consistent with the left-to-right orientation of the thread in these patterns. In radial stitching (Pattern

19) and concentric stitching (Pattern 20), threads are oriented in multiple directions, leading to varied responses. In cross-stitching (Pattern 21), threads are aligned diagonally across the square, but responses were predominantly associated with the top-right to bottom-left direction, where the threads were more prominent. These results suggest that tactile cues at the meso-scale (200 μ m to 1 mm) created by thread alignment play a decisive role in guiding finger movement. Conversely, altering stitching to create grooves or surface textures at the macro-scale (1 mm to 5 cm) did not impact movement guidance.

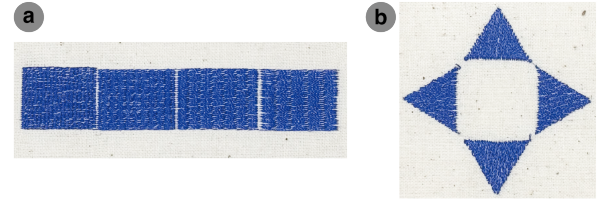


Figure 3: Examples of eyes-free interfaces utilizing tactile properties of stitching styles. (a) Slider UI. (b) Arrow key.

4 The CHI2025 interactivity experience

Altogether, upon reviewing the results of the preliminary study, we are convinced that these patterns will interest the CHI 2025 audience and stimulate engaging discussions. In the following, we detail interactivity use cases that the audience will experience during the conference. After welcoming CHI attendees to our booth, we will begin by simply presenting our idea and approach to attendees. We will expose the 26 tested patterns, alongside others, so the audience can touch them and experience the different tactile properties. We will exchange on the preliminary results of our study and discuss around possible use cases for our embroidery designs as shown in the subsequent sections.

Fabric-Based Eyes-Free Interfaces— Differences in stitching styles can facilitate tactile perception and finger guidance, enabling the design of interfaces that support eyes-free interaction. For example, in a slider UI, commonly used for adjusting volume or brightness, embroidery patterns such as those shown in Figure 3 (a) can guide the finger along the intended direction and provide tactile cues for increasing or decreasing values without visual feedback. Additionally, in a cross-shaped directional keypad, as shown in Figure 3 (b), stitching patterns can indicate the direction of each button, allowing users to identify their current position and direction without visual assistance. Therefore, at the conference, attendees will perform an eyes-free selection task by touching the tactile slider UIs illustrated Figure 4 (a).

Embroidery Design Assistance— This interface enables novice users to understand how design stitching patterns may result to desired textures. In general embroidery design, designers directly adjust various stitching parameters, such as style, deviation and so on, to create patterns that result in specific textures. This knowledge is acquired with practice and experience, and it is therefore challenging for beginners to intuitively imagine how specific stitching parameters result in particular textures. Leveraging the findings

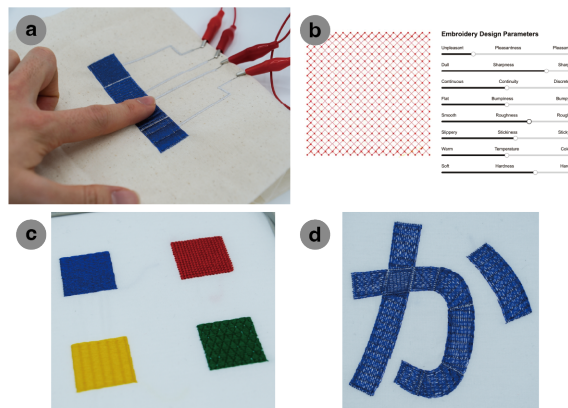


Figure 4: ‘Demonstration of embroidered tactile. (a) Fabric-Based Eyes-Free Interfaces. Different stitch lengths allow users to tactilely feel changes in the slider value. (b) Embroidery Design Assistance. Users can intuitively design embroidery from familiar tactile words. (c) Clothing Management Support for Visually Impaired Individuals. Embroidery allows users to identify differences in color or features of clothing without visual inspection. (d) Sandpaper letters. Tactile embroidery showing the flow of letter strokes helps young children learn to write letters.

of this study, interfaces can enable beginners to design stitching styles using high level parameters (Figure 4 (b)), such as flat vs. bumpy or smooth vs. rough, and use the corresponding stitching parameters to produce embroideries with the desired effects. Attendees will explore embroidery stitches intuitively using our system at the conference, and for some, we will embroider the patterns they envision directly during the interactivity session on the with a sewing machine.

Clothing Management Support for Visually Impaired Individuals— People with visual impairments often face difficulties to identify clothing types, colors or patterns [5]. The embroidered patterns we present could encode information—such as clothing color, pattern, or tactile properties like warmth or coolness (Figure 4 (c)). This approach can support faster and more efficient clothing recognition, enhancing their daily fashion activities. Conference attendees will experience the different embroidered clothing, depending on the color and characteristics of the clothing.

Sandpaper letters (Tactile Letter Learning for Children)— Tactile learning tools, such as Montessori’s sandpaper letters, enable children to trace rough-textured letters as part of their learning process. While existing sandpaper letters allow only for tactile exploration, embroidery can introduce enhanced features like finger guidance and frictional feedback (Figure 4 (d)). These enhancements can help children focus on stroke directions, writing order, and the nuances of writing, such as strokes and hooks, providing a more comprehensive learning experience. Attendees will experience a demonstration in which they can feel the flow of the character strokes from tactile sensations by touching the embroidery of hiragana (Japanese characters). We made the demo shown in the

figure 4 (d) with the stitching method that strongly induces finger movement as revealed in the user study.

5 Conclusion

We presented a method to achieve desired tactile properties on fabric without altering its appearance by varying embroidery stitching techniques. Through an investigation of embroidery design software and expert interviews, we explored stitching styles that influence tactile perception. We then evaluated the effects of stitching variations on tactile perception through user experiments. Based on the findings, we developed four applications that utilize differences in tactile properties.

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